2nd Midterm · Unit 4 · Nonlinear systems and stability · 5 points · 45 min · Conceptual / fast

P08 — Two competing species with a competition parameter b: equilibria, Jacobians, the threshold b = 4/3 between coexistence and competitive exclusion

Two species share a habitat. Their populations $x(t) \geq 0$, $y(t) \geq 0$ (in thousands of individuals, time in years) obey

$$x' = x(6 - 2x - y), \qquad y' = y(4 - y - bx),$$

where $b > 0$ is a constant.

(a) Interpret each term of the model. Find all the equilibria as functions of b and the values of b for which there is an equilibrium with both species present ($x > 0$, $y > 0$).

(b) Compute the Jacobian matrix and classify every equilibrium in the closed first quadrant as b varies. Identify the threshold value of b that separates coexistence from competitive exclusion.

(c) Take $b = 1$. Draw (describe) the nullclines, the eigen-directions at each equilibrium and the phase portrait in the first quadrant. What happens in the long run to a population with $x(0) > 0$, $y(0) > 0$?

(d) Take $b = 2$. Answer the same questions. Which species survives?

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns $x(t), y(t)$; independent variable $t \geq 0$; the right-hand side is polynomial, hence C^1, so solutions exist and are unique (Picard). Autonomous, **nonlinear** (quadratic terms), planar.
- The axes are invariant: $x = 0 \Rightarrow x' = 0$, $y = 0 \Rightarrow y' = 0$. By uniqueness no orbit crosses them, so the closed first quadrant is invariant (populations stay non-negative).
- Parameter: $b > 0$, the harm done to y by each unit of x.

Coefficient	Meaning
$6, 4$	intrinsic growth rates of x, y (year^{-1})
$-2x^2, -y^2$	intraspecific competition; carrying capacities $K_x = 3$, $K_y = 4$
$-xy$ in x'	harm to x from encounters with y
$-bxy$ in y'	harm to y from encounters with x

Phase 2. Canonical formulation

$\mathbf{x}' = \mathbf{F}(\mathbf{x})$ with $\mathbf{F}(x,y) = \big(x(6-2x-y),\ y(4-y-bx)\big)^T$. Near an equilibrium $\mathbf{x}_0$, $\mathbf{u} = \mathbf{x} - \mathbf{x}_0$ obeys $\mathbf{u}' \approx D\mathbf{F}(\mathbf{x}_0)\,\mathbf{u}$; when the equilibrium is hyperbolic (no eigenvalue with zero real part) the Hartman-Grobman theorem (Robinson Sec. 32.4.3) says the nonlinear portrait near it is the linear one.

Part (a) · 1 point

Interpretation, equilibria and existence of coexistence

Step 1. Interpretation

Each species grows logistically when alone ($y=0$: $x' = 2x(3-x)$, capacity 3; $x=0$: $y' = y(4-y)$, capacity 4). The cross terms $-xy$ and $-bxy$ are both negative: each species reduces the growth of the other, so the model is one of **competition** (Robinson Sec. 33.1). The larger b, the stronger the pressure of x on y.

Step 2. Equilibria

$x' = 0$ when $x = 0$ or $2x + y = 6$; $y' = 0$ when $y = 0$ or $y + bx = 4$. The four combinations:

- $x = 0, y = 0$: $(0,0)$.
- $x = 0, y + bx = 4$: $(0,4)$.
- $2x + y = 6, y = 0$: $(3,0)$.
- $2x + y = 6$ and $bx + y = 4$: subtracting, $(2-b)x = 2$, so for $b \neq 2$

$$x^* = \frac{2}{2-b}, \qquad y^* = 6 - 2x^* = \frac{6(2-b)-4}{2-b} = \frac{8-6b}{2-b}.$$

Step 3. Positivity

$x^* > 0 \iff b < 2$. Then $2-b>0$ and $y^* > 0 \iff 8-6b > 0 \iff b < \tfrac{4}{3}$. So the coexistence point is in the open quadrant exactly for $0 < b < \tfrac{4}{3}$. At $b = \tfrac{4}{3}$: $x^* = \frac{2}{2/3} = 3$, $y^* = 0$, it coincides with $(3,0)$. For $\tfrac{4}{3} < b < 2$ it lies in $y < 0$ (no biological meaning); for $b = 2$ the two lines are parallel and there is none.

KEY RESULTS

- Equilibria: $(0,0)$, $(3,0)$, $(0,4)$ and, for $b \neq 2$, $\left(\dfrac{2}{2-b},\ \dfrac{8-6b}{2-b}\right)$.
- The coexistence point lies in $x>0, y>0$ if and only if $0 < b < \tfrac{4}{3}$; at $b = \tfrac{4}{3}$ it merges with $(3,0)$.

Part (b) · 2 points

Jacobian and classification as b varies

Step 1. Jacobian

$\frac{\partial}{\partial x}\left[6x-2x^2-xy\right]=6-4x-y$, $\frac{\partial}{\partial y}\left[6x-2x^2-xy\right]=-x$, $\frac{\partial}{\partial x}\left[4y-y^2-bxy\right]=-by$, $\frac{\partial}{\partial y}\left[4y-y^2-bxy\right]=4-2y-bx$.

Step 2. Origin

$D\mathbf{F}(0,0)=\begin{pmatrix}6&0\\0&4\end{pmatrix}$: two positive eigenvalues, **unstable node**, eigenvectors along the axes. Orbits leave tangent to the slow direction, the y axis ($4<6$).

Step 3. The point (0, 4)

$D\mathbf{F}(0,4)=\begin{pmatrix}6-4&0\\-4b&4-8\end{pmatrix}=\begin{pmatrix}2&0\\-4b&-4\end{pmatrix}$. Lower triangular: eigenvalues 2 and -4, a **saddle** for every b. Eigenvector of -4: $(0,1)$, the y axis is the stable manifold. Eigenvector of 2: $(D\mathbf{F}-2I)\mathbf{v}=\mathbf{0}$ gives row 2 $-4bv_1-6v_2=0$, so $\mathbf{v}=(3,-2b)^T$: the unstable manifold enters the quadrant towards the lower right. Meaning: a few individuals of x introduced into the y-only state grow (their per capita rate there is $6-4=2>0$).

Step 4. The point (3, 0)

$D\mathbf{F}(3,0)=\begin{pmatrix}6-12&-3\\0&4-3b\end{pmatrix}=\begin{pmatrix}-6&-3\\0&4-3b\end{pmatrix}$. Upper triangular: eigenvalues -6 (along the x axis) and $4-3b$, which is the per capita growth rate of a few y individuals in the x-only state.

- $b<\frac{4}{3}$: $4-3b>0$, **saddle** (y can invade).
- $b>\frac{4}{3}$: $4-3b<0$, **stable node** (y cannot invade).
- $b=\frac{4}{3}$: one zero eigenvalue, non-hyperbolic: the linearisation decides nothing there (Robinson Sec. 32.4.3). This is where the coexistence point crosses $(3,0)$.

Step 5. Coexistence point

At (x^*,y^*): $6-4x^*-y^*=(6-2x^*-y^*)-2x^*=-2x^*$ and $4-2y^*-bx^*=(4-y^*-bx^*)-y^*=-y^*$, so

$$D\mathbf{F}(x^*,y^*)=\begin{pmatrix}-2x^*&-x^*\\-by^*&-y^*\end{pmatrix}.$$

- $\mathrm{tr}=-(2x^*+y^*)=-6$ (because $2x^*+y^*=6$ on the nullcline).
- $\det=2x^*y^*-bx^*y^*=(2-b)x^*y^*=(2-b)\frac{2}{2-b}\frac{8-6b}{2-b}=\frac{2(8-6b)}{2-b}$, positive for $0<b<\frac{4}{3}$.
- Discriminant: $\mathrm{tr}^2-4\det=(-2x^*-y^*)^2-4(2x^*y^*-bx^*y^*)=(2x^*-y^*)^2+4bx^*y^*>0$.

Real eigenvalues, both negative ($\mathrm{tr}<0,\det>0$): **stable node** for every $0<b<\frac{4}{3}$. (The off-diagonal entries have the same sign, which always forbids a focus in a competition model.)

Step 6. Summary and threshold

b	$(0,0)$	$(0,4)$	$(3,0)$	**coexistence**
$0<b<\frac{4}{3}$	unstable node	saddle	saddle	stable node in the quadrant
$b=\frac{4}{3}$	unstable node	saddle	non-hyperbolic	merged with $(3,0)$
$b>\frac{4}{3}$	unstable node	saddle	stable node	outside the quadrant or none

The threshold is $b_c=\frac{4}{3}$, the value at which y stops being able to invade the x-only state: $4-3b_c=0$, i.e. $b_c=r_y/K_x$ (growth rate of y over the carrying capacity of x).

KEY RESULTS

- $D\mathbf{F}=\begin{pmatrix}6-4x-y & -x\\ -by & 4-2y-bx\end{pmatrix}$.
- $(0,0)$: $\mathrm{diag}(6,4)$, unstable node for all b.
- $(0,4)$: $\begin{pmatrix}2 & 0\\ -4b & -4\end{pmatrix}$, eigenvalues $2,-4$: saddle for all b.
- $(3,0)$: $\begin{pmatrix}-6 & -3\\ 0 & 4-3b\end{pmatrix}$: saddle for $b<\frac{4}{3}$, stable node for $b>\frac{4}{3}$, non-hyperbolic at $b=\frac{4}{3}$.
- Coexistence ($0<b<\frac{4}{3}$): $\begin{pmatrix}-2x^* & -x^*\\ -by^* & -y^*\end{pmatrix}$, $\mathrm{tr}=-6$, $\det=\dfrac{2(8-6b)}{2-b}>0$, discriminant >0: stable node.
- Threshold $b_c=\frac{4}{3}$.

Part (c) · 1 point

Phase portrait and long-time behaviour for b = 1

Step 1. Eigen-data at (2, 2)

$x^*=\frac{2}{1}=2, y^*=\frac{2}{1}=2$. $D\mathbf{F}=\begin{pmatrix}-4 & -2\\ -2 & -2\end{pmatrix}$, $p(\lambda)=(-4-\lambda)(-2-\lambda)-4=\lambda^2+6\lambda+4$, $\lambda=-3\pm\sqrt{5}$. Row 1 of $(D\mathbf{F}-\lambda I)\mathbf{v}=\mathbf{0}$: $(-4-\lambda)v_1-2v_2=0$, $v_2=-\frac{4+\lambda}{2}v_1$. For $\lambda_s=-3+\sqrt{5}$: $v_2=-\frac{1+\sqrt{5}}{2}v_1$, $\mathbf{v}_s=(2,-1-\sqrt{5})$; for $\lambda_f=-3-\sqrt{5}$: $v_2=\frac{\sqrt{5}-1}{2}v_1$, $\mathbf{v}_f=(2,\sqrt{5}-1)$. Orbits reach $(2,2)$ tangent to $\mathbf{v}_s$ (slope $-\frac{1+\sqrt{5}}{2}\approx-1.62$), except the two on the fast line.

Step 2. Saddles

At $(3,0)$, $b=1$: $\begin{pmatrix}-6 & -3\\ 0 & 1\end{pmatrix}$; for $\lambda=1$, row 1: $-7v_1-3v_2=0$, $\mathbf{v}=(3,-7)$. Its branch pointing into the quadrant, along $(-3,7)$, leaves $(3,0)$ upwards to the left and runs to $(2,2)$. At $(0,4)$: $\begin{pmatrix}2 & 0\\ -4 & -4\end{pmatrix}$, unstable direction $(3,-2)$: its branch in the quadrant leaves $(0,4)$ to the lower right and runs to $(2,2)$. The axes are the stable manifolds of the two saddles.

Step 3. Nullclines and regions

L_x: $2x + y = 6$ (where $x' = 0$, vertical arrows), from $(0, 6)$ to $(3, 0)$. L_y: $x + y = 4$ (where $y' = 0$, horizontal arrows), from $(0, 4)$ to $(4, 0)$. They cross at $(2, 2)$.

- Above both lines: $x' < 0, y' < 0$ (towards the lower left).
- Below both lines: $x' > 0, y' > 0$ (towards the upper right).
- Between them for $x < 2$ (e.g. $(1, 3.5)$): $x' > 0, y' < 0$.
- Between them for $x > 2$ (e.g. $(3, 0.5)$): $x' < 0, y' > 0$.

Step 4. Trapping regions and the long run

Take the triangle $R = \{x + y > 4,\ 2x + y < 6,\ x < 2\}$. On its side on L_x, $x' = 0$ and $y' = y(4 - x - y) = y(x - 2) < 0$ (there $y = 6 - 2x$, so $4 - x - y = x - 2$): the field points down, into R. On its side on L_y, $y' = 0$ and $x' = x(6 - 2x - y) = x(2 - x) > 0$: the field points right, into R. So R is positively invariant; inside it x increases and y decreases, both bounded, so they converge, and the limit must be an equilibrium: $(2, 2)$. The other triangle is treated the same way. An orbit starting above or below both lines moves monotonically (lower left or upper right) until it either enters one of the triangles or converges directly to $(2, 2)$. Hence every orbit with $x(0) > 0, y(0) > 0$ tends to $(2, 2)$: the two species **coexist** at 2000 individuals each.

KEY RESULTS

- Coexistence point $(2, 2)$; $D\mathbf{F} = \begin{pmatrix} -4 & -2 \\ -2 & -2 \end{pmatrix}$, $\lambda = -3 \pm \sqrt{5} \approx -0.764,\ -5.236$.
- Slow eigenvector $(2,\ -1 - \sqrt{5})$, fast eigenvector $(2,\ \sqrt{5} - 1)$.
- Saddles: $(3, 0)$ with unstable direction $(3, -7)$ (eigenvalue 1); $(0, 4)$ with unstable direction $(3, -2)$ (eigenvalue 2).
- Every orbit with $x(0) > 0, y(0) > 0$ tends to $(2, 2)$: **coexistence**.

Part (d) · 1 point

Phase portrait and long-time behaviour for b = 2

Step 1. Equilibria and linearisation

$2x + y = 6$ and $2x + y = 4$ never meet, so only $(0, 0)$, $(3, 0)$, $(0, 4)$ remain. At $(3, 0)$, $4 - 3b = -2$: eigenvalues -6 and -2, a **stable node**; for $\lambda = -2$, row 1 of $(D\mathbf{F} + 2I)\mathbf{v} = \mathbf{0}$ is $-4v_1 - 3v_2 = 0$, so $\mathbf{v} = (3, -4)$, the slow direction to which orbits are tangent on arrival. At $(0, 4)$, $\begin{pmatrix} 2 & 0 \\ -8 & -4 \end{pmatrix}$: a saddle, stable manifold the y axis, unstable direction from row 2 $-8v_1 - 6v_2 = 0$, $\mathbf{v} = (3, -4)$.

Step 2. Regions

- $2x + y > 6$: $x' < 0$ and $y' < 0$ (since also $2x + y > 4$).
- $4 < 2x + y < 6$: $x' > 0, y' < 0$ (e.g. $(1.5, 2)$: $x' = 1.5 \cdot 1 > 0, y' = 2 \cdot (-1) < 0$).
- $2x + y < 4$: $x' > 0, y' > 0$.

On $2x + y = 6$ the field is vertical and points down, on $2x + y = 4$ it is horizontal and points right: both point into the strip, which is positively invariant. In the strip x increases and y decreases, both bounded, so the orbit converges to an equilibrium in the closure of the strip with $x > 0$: $(3, 0)$. Orbits outside the strip move monotonically into it or straight to $(3, 0)$.

Step 3. Long run

Every orbit with $x(0) > 0$ (and any $y(0) \geq 0$) tends to $(3, 0)$: species y becomes extinct and x reaches its carrying capacity. Only the orbits on the y axis ($x(0) = 0$) go to $(0, 4)$. The unstable branch of the saddle $(0, 4)$ runs to $(3, 0)$, leaving along $(3, -4)$.

KEY RESULTS

- Equilibria $(0, 0)$, $(3, 0)$, $(0, 4)$ only; nullclines $2x + y = 6$ and $2x + y = 4$, parallel.
- $(3, 0)$: $\begin{pmatrix} -6 & -3 \\ 0 & -2 \end{pmatrix}$, stable node, slow eigenvector $(3, -4)$.
- $(0, 4)$: $\begin{pmatrix} 2 & 0 \\ -8 & -4 \end{pmatrix}$, saddle, unstable eigenvector $(3, -4)$.
- Every orbit with $x(0) > 0$ tends to $(3, 0)$: y dies out, **competitive exclusion**.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $b \to 0^+$: $x^* \to 1$, $y^* \to 4$; y no longer feels x, and x settles at $6 - 2x - 4 = 0$. $b \to \frac{4}{3}^-$: $(x^*, y^*) \to (3, 0)$ and $\det D\mathbf{F} \to 0$, the exchange of stability with $(3, 0)$ (a transcritical bifurcation, **extension**: a bifurcation where two equilibria cross and swap their stability). For every $b > \frac{4}{3}$ the outcome is that of (d).
- **Checks:** $\mathbf{F} = \mathbf{0}$ at each equilibrium; triangular Jacobians read directly; $\operatorname{tr} D\mathbf{F}(x^*, y^*) = -6$ for every b; at $b = 1$, $\lambda_s + \lambda_f = -6$ and $\lambda_s \lambda_f = 9 - 5 = 4 = \det$. Units: each bracket has units of year^{-1} when the coefficients carry year^{-1} and (thousand individuals)$^{-1}$ year^{-1}.
- **Common mistakes in the exam:** (1) keeping the coexistence point for $\frac{4}{3} < b < 2$ although $y^* < 0$; (2) using the full Jacobian formula at (x^*, y^*) without the simplification $6 - 2x^* - y^* = 0$, which hides that $\operatorname{tr} = -6$; (3) concluding stability at $b = \frac{4}{3}$ from the linearisation (zero eigenvalue); (4) predicting the long-time outcome from local stability alone, without the nullcline and trapping-region argument.